



DRILLHOLE RECORD

CONTRACT NO. GE/2022/08

HOLE NO. **DH 4**

SHEET **1** of **3**

PROJECT Ground Investigation - New Territories East, PWP Item No. 9199WC, Water Supply to New Housing Developments along Route Twisk

METHOD **ROTARY**

CO-ORDINATES

TASK ORDER NO. **GE/2022/08.11B**

MACHINE **SD44**

E **829015.24**
N **827503.60**

DATE **17.12.2022** to **22.12.2022**

FLUSHING MEDIUM **WATER**

ORIENTATION **VERTICAL**

GROUND LEVEL **+176.28 mPD**

Drilling Progress	Casing Size	Water Level (m) Shift Start/End	Water Return%	TCR%	SCR%	RQD%	Fracture Index	Tests	Samples	Reduced Level	Depth (m)	Legend	Grade	Description
17.12.2022	SW									0.00	+176.28	0.00		Brown (7.5YR 5/4) mottled yellowish red, slightly clayey SILT with much angular to subangular fine to coarse gravel. (FILL)
1									1	0.45				
	SW 1.50m		85	76					2	0.95	+175.28	1.00		Greyish brown (10YR 5/2) and reddish yellow (7.5YR 8/6), angular to subangular, clayey silty sandy fine to coarse GRAVEL and much cobble of rock fragments. (FILL)
2	PW		80	100					3	1.50	+174.98	1.30		Very stiff, reddish yellow (7.5YR 8/6) mottled brownish yellow, clayey sandy SILT with some angular to subangular fine to coarse gravel. (COLLUVIUM)
3									4	2.50				
									5	2.60				
									6	2.70				
									7	3.00				
									8	3.30				
4			80	83					9	3.50	+172.78	3.50	IV	Weak, light yellowish brown (2.5Y 6/3), highly decomposed, TUFF BRECCIA. (Angular, slightly silty sandy fine to medium GRAVEL of highly decomposed tuff breccia fragments)
17.12.2022		1.17 at 1800							10	4.50				
18.12.2022		4.49 at 0800							11	4.60				
5									12	5.50	+170.78	5.50	IV	Weak, light yellowish brown (2.5Y 6/3), highly decomposed, TUFF BRECCIA. (Angular, slightly silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
6			80	83					13	6.50				
7									14	6.82				
	PW 7.50m								15	7.50				
8	HW		80	80					16	8.25	+167.93	8.35	IV	Weak to moderately weak, reddish yellow (7.5YR 8/6), highly decomposed, TUFF BRECCIA. (Angular, clayey silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
9			80	92	9	0	NI		T2-101	9.00	+167.28	9.00	III	Moderately strong to strong, light brownish grey and light grey, moderately decomposed, TUFF BRECCIA. (CORESTONE)
			80	92	78	68	7.8		T2-101	9.65	+168.77	9.51	V	Extremely weak, pale olive (5Y 6/3), completely decomposed, TUFF BRECCIA. (Stiff, slightly sandy
19.12.2022		0.33 at 1800							17	9.80	+168.38	9.90		
20.12.2022		9.10							T2-101	9.90				

- SMALL DISTURBED SAMPLE

LARGE DISTURBED SAMPLE

U76 SAMPLE

PISTON SAMPLE (76mm)

MAZIER SAMPLE

SPT LINER SAMPLE

WATER SAMPLE

U100 SAMPLE
- STANDARD PENETRATION TEST

IN-SITU VANE SHEAR TEST

PACKER TEST

PERMEABILITY TEST

PRESSUREMETER TEST

BOREHOLE TELEVIEWER

PIEZOMETER TIP

STANDPIPE TIP

LOGGED **C. Chan**

DATE **28.12.2022**

CHECKED **R. Chu**

DATE **03.01.2023**

REMARKS

1. An inspection pit was excavated to 1.00m deep by hand tools.

2. Constant head permeability tests were carried out at sections from 3.10m to 4.60m and from 16.10m to 17.60m.

3. A standpipe was installed to 4.50m.

4. A piezometer was installed with tip at 18.00m.



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11		at 0800	80	100	0	0	>20		T2-101		10.45		III	clayey SILT) Moderately weak to moderately strong, light grey and light yellowish brown, moderately decomposed, TUFF BRECCIA. (CORESTONE)
			80	82	45	23	12.7		T2-101	+165.08	11.20		IV	Weak to moderately weak, yellowish brown (10YR 5/8), highly decomposed, TUFF BRECCIA. (Angular, clayey silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
12			80	96	0	0	NI		T2-101	+164.78	11.50		V	From 11.50m to 11.73m and 13.10m to 13.30m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
13			80	64	0	0	NR		T2-101	+164.55	11.73		IV	
			80	0					T2-101	+163.18	13.10		V	
14			80	86	0	0	NI		T2-101	+162.99	13.30		V	Extremely weak, pale olive (5Y 6/3), completely decomposed, TUFF BRECCIA. (Stiff, clayey sandy SILT)
			80	86	0	0	NR		T2-101	+162.64	13.64		IV	Weak to moderately weak, olive grey 5Y 5/2), highly decomposed, TUFF BRECCIA. (Angular, clayey silty sandy fine to coarse GRAVEL of highly decomposed tuff breccia fragments)
15			80	82					T2-101	+161.90	14.38		V	From 14.38m to 14.50m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
16									T2-101	+161.78	14.50		IV	From 17.60m to 17.73m: Moderately strong and moderately decomposed. (CORESTONE)
17			80	60					T2-101		15.50			
18		8.91 at 1800	80	100	15	0	NI		T2-101		15.91			
19		16.70 at 0800	80	96	63	63	7.4		T2-101	+158.68	17.60		III	Moderately strong, light grey dappled light yellowish brown, moderately decomposed, TUFF BRECCIA. (CORESTONE)
20									T2-101	+158.55	17.73		IV	From 18.98m to 19.48m: Strong and slightly decomposed.
									T2-101	+157.30	18.98		III	From 19.93m to 20.22m: No recovery, assumed to
									T2-101	+156.80	19.48		III	
									T2-101	+156.35	19.93			

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	HW 20.43m		80	65	57	42	NR		T2-101	+158.08	20.22		V	be completely decomposed TUFF BRECCIA.
			80	100	100	100	5.3		T2-101	+155.68	20.60		III	Strong, light grey dappled light yellowish brown and grey, slightly decomposed, TUFF BRECCIA.
21			80	100	100	100			T2-101				II	Joints are medium to widely spaced, occasionally closely spaced, rough planar and rough stepped, iron oxide stained, chlorite and calcite coated, dipping at 0° to 10°, 10° to 20° and subvertically from 24.66m to 24.76m.
22		6.32 at 1800	80	100	100	100			T2-101					From 20.22m to 20.60m and 25.24m to 25.42m: Moderately strong to strong and moderately decomposed.
22.12.2022		17.15 at 0800	80	100	100	100	0.2		T2-101					From 25.14m to 25.24m: No recovery, assumed to be completely decomposed TUFF BRECCIA.
23			80	100	100	100			T2-101					
24			80	100	100	100			T2-101					
25			80	100	100	100			T2-101					
25			80	93	82	82	5.0		T2-101	+151.14	25.06		V	
25							NR		T2-101	+151.04	25.14		V	
25									T2-101	+150.88	25.24		III	
25							2.5		T2-101	+150.88	25.42		III	
25									T2-101	+150.25	26.03		II	
26		17.58 at 1800												End of hole at 26.03 m.
27														
28														
29														
30														

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LARGE DISTURBED SAMPLE

U70 SAMPLE

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